

Mini Operating System Simulator

Project Report

Name	Shaid Ibna Sobhan
Student ID	2102057
Registration No.	10184
Course Code	CIT 322
Course Title	Operating System Sessional
Project Title	Mini Operating System Simulator using Python Tkinter

Project Overview

The Mini Operating System Simulator is a GUI-based educational application developed using Python and Tkinter. The primary objective of this project is to demonstrate the working principles of fundamental Operating System algorithms through interactive simulations. Users can provide input, execute different algorithms, observe step-by-step execution, and compare the generated results. The simulator enhances conceptual understanding by combining theoretical knowledge with practical visualization. All modules are connected through a single, colorful home dashboard, allowing the user to navigate freely between topics while each module continues to function independently.

Implemented Modules

The simulator consists of the following five modules:

1. CPU Scheduling

- ▶ First Come First Serve (FCFS)
- ▶ Shortest Job First (SJF – Non-Preemptive)
- ▶ Round Robin (RR)
- ▶ Priority Scheduling (Non-Preemptive)
- ▶ Multilevel Queue Scheduling

2. Process and Thread Management

- ▶ Process State Lifecycle (New, Ready, Running, Waiting, Terminated)
- ▶ Step-by-Step State Transition Visualization
- ▶ Process vs Thread Comparison

3. Process Synchronization

- ▶ Producer-Consumer Problem (Bounded Buffer)
- ▶ Semaphore-based Implementation (mutex, empty, full)
- ▶ Real Multi-threaded Execution
- ▶ Live Buffer and Event Log Visualization

4. Deadlock Avoidance

- ▶ Banker's Algorithm
- ▶ Allocation, Maximum and Need Matrix
- ▶ Safe Sequence Detection
- ▶ Safe / Unsafe State Identification

5. Memory Management

- ▶ Contiguous Allocation: First Fit, Best Fit, Worst Fit
- ▶ Paging: Page Table Generation
- ▶ Internal Fragmentation Calculation

Main Features

- Interactive Graphical User Interface (GUI)
- Colorful, Card-based Home Dashboard
- User-friendly, Editable Input Tables
- Live Gantt Chart Visualization for CPU Scheduling
- Dynamic, Algorithm-specific Explanation Boxes
- Safe Sequence Calculation (Banker's Algorithm)
- Resource Allocation Monitoring
- Real Thread-based Synchronization Demonstration
- Home and Back Navigation on Every Module
- Error Handling and Input Validation
- Modular Python Programming Structure
- Separate Python File for Each Module

Technologies Used

Component	Technology
Programming Language	Python 3
GUI Framework	Tkinter
Visualization Library	Matplotlib
Programming Paradigm	Object-Oriented Programming
Project Structure	Modular Programming (one file per module)
Operating System (Tested)	Manjaro Linux
Development Environment	Visual Studio Code / PyCharm

Learning Outcomes

This project provides practical knowledge of several important Operating System concepts including:

- CPU Scheduling Techniques
- Process and Thread State Management
- Process Synchronization using Semaphores
- Deadlock Avoidance using the Banker's Algorithm
- Memory Allocation Strategies

- Concurrent Process Execution
- GUI-based Simulation Development
- Modular Software Design in Python

Conclusion

The Mini Operating System Simulator provides an effective platform for understanding core Operating System algorithms through practical implementation and visualization. The integration of CPU scheduling, process and thread management, synchronization, deadlock avoidance, and memory management modules enables students to observe how different Operating System components work in real-world scenarios. The project successfully combines theoretical concepts with graphical simulation, making the learning process more interactive, efficient, and user-friendly.